Forgetting and Retention in Language Model Domain Adaptation under a Fixed Replay Budget

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Abstract

Domain adaptation can improve a language model on new text while degrading its performance on earlier distributions. This project will investigate how replay sources, replay objectives, and sample selection affect this trade-off under a fixed replay budget. A controlled study will compare substitute general-domain text and self-generated text, cross-entropy and teacher–student KL constraints, and random, stratified, and cluster-based selection. Experiments will use Pythia-410M, mathematical domain text, and independent general-domain evaluation sets. Unique replay coverage, cumulative replay exposure, and actual computation will be recorded separately. The intended contribution is a reproducible comparison and an analysis of when lightweight selection helps or fails. Main experiments are planned for eight weeks, followed by thesis preparation. No training results are claimed in this proposal.

1 Background and Related Work

Continued training on a specialized corpus is a practical way to adapt a pretrained language model. However, updates that fit the new distribution may interfere with capabilities supported by its existing parameters. This resembles catastrophic forgetting in computer vision continual learning: both face a stability–plasticity trade-off and employ parameter constraints, output distillation, or replay. Elastic weight consolidation, Learning without Forgetting, and deep generative replay provide foundational examples [5, 6, 7]. Here, *domain adaptation* means continued next-token prediction training, and *forgetting* means unintended performance degradation, distinct from deliberate machine unlearning.

Recent language-model studies examine different aspects of this problem. Chang et al. [1] inject fictional facts during pretraining and track acquisition and subsequent decay using answer-span log probabilities. Such probes measure controlled knowledge retention; they are not equivalent to free-generation accuracy. Bethune et al. [2] study pretraining-data injection during fine-tuning. Small injection rates, around 1% in their settings, substantially mitigate forgetting, while repeatedly using a small replay bank can cause overfitting. These rates should not be assumed optimal for a different model or domain. Marek et al. [3] show that self-generated replay can substitute for original data in some settings, with KL constraints often more stable than cross-entropy, but outcomes depend on capacity and optimization.

Replay selection is already an established idea. Self-Synthesized Rehearsal (SSR) combines instruction synthesis, output refinement, and representative selection through clustering [8]. Kotha and Liang [9] show that replay can also improve target performance, so its effect cannot be described solely as sacrificing adaptation for retention. Wu et al. [4] explain benefits of data mixtures under specific high-dimensional regression assumptions; this does not guarantee retention in language models. Broader surveys and empirical continual fine-tuning studies further motivate explicit evaluation choices [10, 11].

2 Research Questions and Scope

The central question is: **How do replay source, objective, and sample selection influence domain adaptation and retention when replay resources are limited?**

1. **Source and objective:** How do substitute general-domain and self-generated replay compare under cross-entropy (CE) and a frozen-teacher KL constraint?
2. **Selection:** At equal replay-bank size, do teacher-NLL stratification or cluster representatives improve the retention–adaptation trade-off relative to random selection?
3. **Budget and cost:** How sensitive are results to cumulative replay exposure, and do any benefits justify generation, teacher inference, and selection costs?

The contribution will be a controlled empirical comparison, rather than a claim that generated replay, KL distillation, or clustering is novel. A plausible hypothesis is that selection reduces finite-bank variability; another is that KL improves retention by preserving the teacher's soft predictions. Both may fail when generated text poorly covers evaluation distributions. The required scope is one main model, one target domain, and three seeds. Full pretraining, broad scaling-law fitting, and a new general-purpose continual-learning framework are excluded.

3 Methodology

3.1 Model and Data

The main model will be the final checkpoint of Pythia-410M [12], with a 160M model available for debugging. Full-parameter continued training will use English mathematical text from FineMath's `finemath-4plus` subset. C4 English will supply substitute general-domain replay. Held-out C4 documents and WikiText-103 will provide separate general-domain evaluations. C4 is not claimed to be Pythia's actual pretraining corpus, and mathematical text likelihood is not a direct measure of mathematical reasoning.

Public model and dataset cards have been checked. FineMath provides only a training split, so document-level training, validation, and test partitions must be constructed before token chunking. General-domain evaluation documents will be disjoint from replay, with cross-split duplicate checks. Versions, document identifiers, tokenization, and subset construction will be recorded. Budgets will be measured using the model's tokenizer rather than dataset-card token counts.

3.2 Replay Objectives

Let θ_0 denote the frozen initial model and D_R a replay bank. The target-domain objective $\mathcal{L}_{\text{new}}$ is next-token negative log-likelihood (NLL). Replay uses either token CE or forward KL from teacher to student:

$$\mathcal{L}_{\text{KL}} = \mathbb{E}_{x\sim D_R}\frac{1}{T_x}\sum_{t=1}^{T_x} D_{\text{KL}}[p_{\theta_0}(\cdot \mid x_{<t}) \,\|\, p_\theta(\cdot \mid x_{<t})]\,, \qquad \mathcal{L} = \mathcal{L}_{\text{new}} + \lambda\mathcal{L}_R. \tag{1}$$

CE uses the actual replay tokens; KL uses the teacher's predictive distribution. Each loss will be normalized over valid tokens. The teacher remains fixed, and loss weight λ is recorded separately from the data budget. Relations between generated-token CE and KL require assumptions about the sampling distribution; equivalence will not be assumed for filtered samples.

3.3 Controlled Comparisons

Table 1 defines eight groups. B0–B4 isolate the source–objective comparison; M, C, and D examine selection and deduplication. Generated text will come from the initial model under recorded prompts, sampling settings, and seeds. Generated-replay methods within each seed will share the same candidate pool.

Group	Replay source and selection	Objective / role
B0	None	Target CE; forgetting baseline
B1 / B2	Substitute general-domain text	Replay CE / KL
B3 / B4	Self-generated, random selection	Replay CE / KL
M	Self-generated, deduplicated and stratified	KL; lightweight selection
C	Self-generated, cluster representatives	KL; established selection idea
D	Self-generated, deduplicated and random	KL; deduplication ablation

Table 1: Planned groups. B1/B2 and B3/B4 are separate runs.

M will deduplicate equal-length chunks, divide them into five equal-frequency strata by teacher mean NLL, and sample proportionally. This does not necessarily change the marginal NLL distribution; it tests finite-bank representation and variability. D separates the effect of deduplication from stratification. C will select representatives using frozen-teacher chunk features and clustering. It is an SSR-inspired selection control, not a replication of the complete SSR instruction framework. Candidate pools and selected token budgets will be matched across selection comparisons; duplicate rates, feature extraction, and selection costs will be logged. Test data will never guide selection.

4 Evaluation, Budgets, and Feasibility

4.1 Retention and Adaptation

For the adapted checkpoint θ_{ft}, the primary quantities are

$$F_{\text{old}} = L_{\text{old}}(\theta_{\text{ft}}) - L_{\text{old}}(\theta_0), \qquad G_{\text{new}} = L_{\text{new}}(\theta_0) - L_{\text{new}}(\theta_{\text{ft}}). \tag{2}$$

Larger F_{old} indicates greater degradation on a specified old distribution; larger G_{new} indicates better target adaptation. Results for C4 and WikiText-103 will be reported separately. Teacher–student KL on held-out data will serve as an output-drift diagnostic, not a replacement for performance evaluation.

Methods will be compared at equal target-token exposure and, where feasible, within a common achieved target-validation-loss range. This helps distinguish effective retention from simply learning the new domain less. Checkpoints and hyperparameters will be selected using validation data; final results will use independent tests. Three seeds will yield means, standard deviations, individual runs, and retention–adaptation trajectories. Conclusions will be limited to the evaluated distributions, without claiming retention of all pretrained knowledge.

4.2 Resource Accounting and Risk Control

Unique replay-bank size K will be distinguished from cumulative replay ratio $b = T_R/T_{\text{new}}$, where T_R and T_{new} count processed replay and target tokens. The fraction of replay among all processed tokens is $p = b/(1+b)$. Initial settings are approximately 2M unique target tokens, 10M target-token exposures, a 1M-token generated candidate pool, a 0.1M-token replay bank, and $b = 0.05$. Pilots will check feasibility and then freeze these settings before formal comparisons.

Eight groups with three seeds give 24 main runs. Additional $b = 0.01$ and 0.10 runs for B4 and M add 12, for 36 planned formal runs. Pilot and tuning runs will be counted separately. A small shared learning-rate and λ search will be specified in advance, with equal tuning allowances for comparable methods. Target training schedules and preprocessing will be held fixed where applicable.

Four RTX 5090 GPUs are available for independent runs; throughput and memory use will be measured rather than inferred from GPU count. Costs will include generation, teacher inference, feature extraction, selection, training, and evaluation. Equal token budgets will not be described as equal compute. The five core baselines take priority if the schedule becomes tight. Main risks are weak generated coverage, mismatch between substitute replay and old evaluations, or no gain from selection. Negative results will still support useful ablations and failure-condition analysis. A second domain, a roughly 1B model, or LoRA [13] is optional and outside the required scope.

5 Schedule and Expected Outcomes

The main study is planned for six to eight weeks within a three-month thesis schedule. The one-day proposal deadline covers literature synthesis and planning, not completed experiments.

Period	Milestone
Weeks 1–2	Complete core reading; fix splits; validate evaluation and short runs; measure costs.
Weeks 3–4	Run source–objective baselines; plot trajectories; draft methods.
Weeks 5–6	Evaluate selection, deduplication, and replay-budget sensitivity.
Weeks 7–8	Complete three seeds; finalize figures and freeze main experiments.
Weeks 9–12	Write and revise the thesis; prepare defense; add only necessary checks.

Deliverables will include reproducible code and configurations, split and budget records, retention–adaptation plots, a cost comparison, and the undergraduate thesis. Six core or closely related papers and public resource cards have been collected; model weights and training subsets have not yet been validated through training. The goal is a defensible answer to the research questions, with no prior promise of improvement or publication.

Declaration of AI usage

During the preparation of this work, the author used ChatGPT to assist with literature research and report writing. After using AI, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

[1] H. Chang, J. Park, S. Ye, et al. How Do Large Language Models Acquire Factual Knowledge During Pretraining? NeurIPS, 2024. arXiv:2406.11813.

[2] L. Bethune, D. Grangier, D. Busbridge, et al. Scaling Laws for Forgetting during Finetuning with Pretraining Data Injection. ICML, PMLR 267, 2025. arXiv:2502.06042.

[3] M. Marek, D. Cho, S. Qiu, et al. Forgetting in Language Models: Capacity, Optimization, and Self-Generated Replay. Preprint, 2026. arXiv:2605.26097.

[4] D. Wu, L. Chen, T. Misiakiewicz, and M. Mondelli. When do data mixtures improve scaling laws? Insights from high-dimensional regression. Preprint, 2026. arXiv:2609.38011.

[5] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. PNAS, 114(13):3521–3526, 2017. doi:10.1073/pnas.1611835114.

[6] Z. Li and D. Hoiem. Learning without Forgetting. Preprint, 2016. arXiv:1606.09282.

[7] H. Shin, J. K. Lee, J. Kim, and J. Kim. Continual Learning with Deep Generative Replay. NeurIPS, 2017. arXiv:1705.08690.

[8] J. Huang, L. Cui, A. Wang, et al. Mitigating Catastrophic Forgetting in Large Language Models with Self-Synthesized Rehearsal. Preprint, 2024. arXiv:2403.01244.

[9] S. Kotha and P. Liang. Replaying pre-training data improves fine-tuning. Preprint, 2026. arXiv:2603.04964.

[10] Z. Wang, E. Yang, L. Shen, and H. Huang. A Comprehensive Survey of Forgetting in Deep Learning Beyond Continual Learning. Preprint, 2023. arXiv:2307.09218.

[11] Y. Luo, Z. Yang, F. Meng, et al. An Empirical Study of Catastrophic Forgetting in Large Language Models During Continual Fine-tuning. Preprint, 2023. arXiv:2308.08747.

[12] S. Biderman, H. Schoelkopf, Q. Anthony, et al. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. Preprint, 2023. arXiv:2304.01373.

[13] E. J. Hu, Y. Shen, P. Wallis, et al. LoRA: Low-Rank Adaptation of Large Language Models. Preprint, 2021. arXiv:2106.09685.